

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 2

Name:

Student ID:

Due on Oct. 7, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

Show that for any events A and B ,

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B).$$

For each of these three inequalities, give a simple criterion for when the inequality is actually an equality (e.g., give a simple condition such that $P(A \cap B) = P(A \cup B)$ if and only if the condition holds).

Problem 2

Let A and B be events. The *symmetric difference* $A\triangle B$ is defined to be the set of all elements that are in A or B but not both. In logic and engineering, this event is also called the XOR (*exclusive or*) of A and B . Show that

$$P(A\triangle B) = P(A) + P(B) - 2P(A \cap B),$$

directly using the axioms of probability.

Problem 3

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$.

1. Give an example of independent events A and B in a finite sample space S (with neither equal to $\emptyset$ or S).
2. Consider the experiment of picking a random point in the rectangle

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\},$$

where the probability of the point being in any particular region contained within R is the area of that region. Let A_1 and B_1 be rectangles contained within R , with areas not equal to 0 or 1. Let A be the event that the random point is in A_1 , and B be the event that the random point is in B_1 . Give a geometric description of when it is true that A and B are independent. Also, give an example where they are independent and another example where they are not independent.

3. Show that if A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c).$$

Problem 4

Consider four nonstandard dice (the *Efron dice*), whose sides are labeled as follows (the 6 sides on each die are equally likely).

A: 4, 4, 4, 4, 0, 0

B: 3, 3, 3, 3, 3, 3

C: 6, 6, 2, 2, 2, 2

D: 5, 5, 5, 1, 1, 1

These four dice are each rolled once. Let A be the result for die A , B be the result for die B , etc.

1. Find $P(A > B)$, $P(B > C)$, $P(C > D)$, and $P(D > A)$.
2. Is the event $A > B$ independent of the event $B > C$? Is the event $B > C$ independent of the event $C > D$? Explain.